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Course:Programming in Java
Course code: CSA0974

1. Hospital Patient Billing System
Develop a Java program for a hospital billing system using Class and Objects.
Data Members:
• Patient ID
• Patient Name
• Room Type (General/Special)
• Number of Days Admitted
• Treatment Charges
Methods:
1. Read patient details.
2. Calculate room charges based on room type.
3. Calculate the total hospital bill.
4. Display patient billing details.
Note: General Room = ₹1,500/day, Special Room = ₹3,000/day.

CODE:

```
import java.util.Scanner;
public class HospitalBill {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int patientId, days;
        String patientName, roomType;
        double treatmentCharges, roomCharges, totalBill;
        System.out.print("Enter Patient ID: ");
        patientId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Patient Name: ");
        patientName = sc.nextLine();
        System.out.print("Enter Room Type (General/Special): ");
        roomType = sc.nextLine();
        System.out.print("Enter Number of Days: ");
        days = sc.nextInt();
        System.out.print("Enter Treatment Charges: ");
        treatmentCharges = sc.nextDouble();
        if (roomType.equalsIgnoreCase("General"))
            roomCharges = days * 1500;
        else
            roomCharges = days * 3000;
        totalBill = roomCharges + treatmentCharges;
        System.out.println("\nPatient ID: " + patientId);
        System.out.println("Patient Name: " + patientName);
        System.out.println("Room Charges: ₹" + roomCharges);
```

```
        System.out.println("Treatment Charges: ₹" + treatmentCharges);
        System.out.println("Total Bill: ₹" + totalBill);
    }
}
```

INPUT:
Enter Patient ID: 101
Enter Patient Name: Ravi
Enter Room Type (General/Special): General
Enter Number of Days: 4
Enter Treatment Charges: 5000

**OUTPUT:**
Patient ID: 101
Patient Name: Ravi
Room Charges: ₹6000.0
Treatment Charges: ₹5000.0
Total Bill: ₹11000.0

2. Airline Ticket Reservation System
Develop a Java program for an airline reservation system using Class and Objects.

**Data Members:**
- Passenger ID
- Passenger Name
- Flight Number
- Ticket Fare
- Number of Tickets

**Methods:**
1. Read passenger details.
2. Calculate total ticket cost.
3. Apply a 5% discount if more than 5 tickets are booked.
4. Display booking and payment details.

Note: GST = 12% on the final amount.

CODE:
```
import java.util.Scanner;
public class AirlineReservation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int passengerId, tickets;
        String passengerName, flightNo;
        double fare, total, discount = 0, gst, finalAmount;
        System.out.print("Enter Passenger ID: ");
        passengerId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Passenger Name: ");
        passengerName = sc.nextLine();
        System.out.print("Enter Flight Number: ");
```

```
        flightNo = sc.nextLine();
        System.out.print("Enter Ticket Fare: ");
        fare = sc.nextDouble();
        System.out.print("Enter Number of Tickets: ");
        tickets = sc.nextInt();
        total = fare * tickets;
        if (tickets > 5) {
            discount = total * 0.05;
        }
        total = total - discount;
        gst = total * 0.12;
        finalAmount = total + gst;
        System.out.println("\nPassenger ID: " + passengerId);
        System.out.println("Passenger Name: " + passengerName);
        System.out.println("Flight Number: " + flightNo);
        System.out.println("Ticket Cost: ₹" + (fare * tickets));
        System.out.println("Discount: ₹" + discount);
        System.out.println("GST (12%): ₹" + gst);
        System.out.println("Final Amount: ₹" + finalAmount);
    }
}
```

## INPUT:

Enter Passenger ID: 103
Enter Passenger Name: Karthik
Enter Flight Number: AI789
Enter Ticket Fare: 3500
Enter Number of Tickets: 8

## OUTPUT:

Passenger ID: 103
Passenger Name: Karthik
Flight Number: AI789
Ticket Cost: ₹28000.0
Discount: ₹1400.0
GST (12%): ₹3192.0
Final Amount: ₹29792.0

## 3. Vehicle Rental Management System

## Develop a Java program for a vehicle rental company using Class and Objects.

**Data Members:**

- Customer ID
- Customer Name
- Vehicle Type (Car/Bike)
- Number of Rental Days

**Methods:**

1. Read customer details.
2. Calculate rental charges.

3. Add a late return penalty if applicable.
4. Display the final rental bill.
Note: Car = ₹2,000/day, Bike = ₹500/day.

## CODE:

```
import java.util.Scanner;
public class VehicleRental {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int customerId, days, penalty;
        String customerName, vehicleType;
        double rent, totalBill;
        System.out.print("Enter Customer ID: ");
        customerId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Customer Name: ");
        customerName = sc.nextLine();
        System.out.print("Enter Vehicle Type (Car/Bike): ");
        vehicleType = sc.nextLine();
        System.out.print("Enter Rental Days: ");
        days = sc.nextInt();
        System.out.print("Enter Late Return Penalty (0 or 500): ");
        penalty = sc.nextInt();
        if (vehicleType.equalsIgnoreCase("Car"))
            rent = days * 2000;
        else
            rent = days * 500;
        totalBill = rent + penalty;
        System.out.println("\nCustomer ID: " + customerId);
        System.out.println("Customer Name: " + customerName);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Rental Charges: ₹" + rent);
        System.out.println("Penalty: ₹" + penalty);
        System.out.println("Final Bill: ₹" + totalBill);
    }
}
```

### INPUT:

Enter Customer ID: 101
Enter Customer Name: Arun
Enter Vehicle Type (Car/Bike): Car
Enter Rental Days: 3
Enter Late Return Penalty (0 or 500): 500

OUTPUT:

Customer ID: 101
Customer Name: Arun

Vehicle Type: Car
Rental Charges: ₹6000.0
Penalty: ₹500
Final Bill: ₹6500.0

## 4. Manufacturing Production Monitoring System
## Develop a Java program to monitor daily production in a manufacturing company
## using Class and Objects.

**Data Members:**
- Product ID
- Product Name
- Target Quantity
- Produced Quantity

**Methods:**
1. Read product details.
2. Compare actual production with the target.
3. Calculate production efficiency percentage.
4. Display production status (Achieved/Not Achieved).

Note: Efficiency = (Produced Quantity / Target Quantity) × 100.

CODE:
```
import java.util.Scanner;
public class ProductionMonitor {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int productId, targetQty, producedQty;
        String productName;
        double efficiency;
        System.out.print("Enter Product ID: ");
        productId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Product Name: ");
        productName = sc.nextLine();
        System.out.print("Enter Target Quantity: ");
        targetQty = sc.nextInt();
        System.out.print("Enter Produced Quantity: ");
        producedQty = sc.nextInt();
        efficiency = (double) producedQty / targetQty * 100;
        System.out.println("\nProduct ID: " + productId);
        System.out.println("Product Name: " + productName);
        System.out.println("Efficiency: " + efficiency + "%");
        if (producedQty >= targetQty)
            System.out.println("Status: Achieved");
        else
            System.out.println("Status: Not Achieved");
    }
}
```

INPUT:
Enter Product ID: 101
Enter Product Name: Fan
Enter Target Quantity: 1000
Enter Produced Quantity: 1200

OUTPUT:
Product ID: 101
Product Name: Fan
Efficiency: 120.0%
Status: Achieved

## 5. Online Course Enrollment System
## Develop a Java program for an online learning platform using Class and Objects.

Data Members:
- Student ID
- Student Name
- Course Name
- Course Fee

Methods:
1. Read student and course details.
2. Calculate the payable amount.
3. Provide a 15% scholarship discount if the fee exceeds ₹20,000.
4. Display enrollment and payment details.

Note: Registration fee = ₹500.

CODE:
```
import java.util.Scanner;
public class CourseEnrollment {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int studentId;
        String studentName, courseName;
        double courseFee, discount = 0, payableAmount;
        System.out.print("Enter Student ID: ");
        studentId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();
        System.out.print("Enter Course Name: ");
        courseName = sc.nextLine();
        System.out.print("Enter Course Fee: ");
        courseFee = sc.nextDouble();
        if (courseFee > 20000) {
            discount = courseFee * 0.15;
        }

        payableAmount = courseFee - discount + 500;
        System.out.println("\nStudent ID: " + studentId);
```

```
        System.out.println("Student Name: " + studentName);
        System.out.println("Course Name: " + courseName);
        System.out.println("Course Fee: ₹" + courseFee);
        System.out.println("Scholarship Discount: ₹" + discount);
        System.out.println("Registration Fee: ₹500");
        System.out.println("Payable Amount: ₹" + payableAmount);
    }
}
```

## INPUT:

Enter Student ID: 101
Enter Student Name: Arun
Enter Course Name: Java Programming
Enter Course Fee: 25000

## OUTPUT:

Student ID: 101
Student Name: Arun
Course Name: Java Programming
Course Fee: ₹25000.0
Scholarship Discount: ₹3750.0
Registration Fee: ₹500
Payable Amount: ₹21750.0